

Modified GBM · model specification

Markov direction · split-normal magnitudes · discrete log-price recursion

What this model is

- Modified GBM is a discrete-time replacement for geometric Brownian motion. Standard GBM draws each log-return from one normal $N(\mu\Delta t, \sigma^2\Delta t)$. That forces a single drift, a single volatility, and independent signs from one bar to the next.
- The replacement keeps the GBM price map $S_{t+1} = S_t \exp(r_t)$, but splits r_t into a direction and a magnitude. Direction is a two-state Markov chain (up / down). Magnitude is drawn from a different truncated-normal law on up bars than on down bars.
- The model is return-based. Parameters are estimated from lookback log-returns only. Listed option quotes are not used in calibration. American calls are priced afterwards by Longstaff-Schwartz LSM on risk-neutral paths.

Why it exists

- Equity returns cluster in sign: up days tend to follow up days, and down days follow down days, more often than an i.i.d. sign would imply.
- Up and down moves need not have the same typical size. A single σ cannot represent that split.
- GBM, GARCH, Heston, and Merton already sit in the V3 comparison. Modified GBM is the extra return-based specification that carries those two empirical facts without adding stochastic variance or Poisson jumps.

What it is not

- Not Black-Scholes GBM with a patched drift. The sign process is Markov, not independent.
- Not a regime-switching GBM with hidden states. The state is the previous bar's observed sign, which is known at t .
- Not Heston: variance is not a mean-reverting diffusion. Not Merton: there is no jump intensity. Not GARCH: there is no ω, α, β recursion on σ^2_t .

Modified GBM · three stages

Every simulated bar uses the same three-stage map.

Stage 1 — direction

$$U = \{r_t > 0\}, \quad D = \{r_t < 0\}$$

- The previous non-zero return's sign selects the next-bar probability. After an up bar the next bar is up with $P(U|U)$ and down with $P(D|U)=1-P(U|U)$. After a down bar the pair is $P(U|D)$, $P(D|D)$.
- High $P(U|U)$ or $P(D|D)$ produces runs of the same sign. High $P(U|D)$ or $P(D|U)$ produces reversals.
- Zero returns are dropped from the transition count. The last observed sign in the lookback, `last_up`, starts the Monte Carlo chain.

Stage 2 — magnitude

$$m_t|U \sim |N(\mu_U, \sigma_U^2)|, \quad m_t|D \sim |N(\mu_D, \sigma_D^2)|$$

- μ_U and σ_U are the sample mean and sample SD of $|r|$ on up bars in the lookback. μ_D and σ_D are the same statistics on down bars. Units are the bar's log-return (daily in the 1.5-year study).
- The absolute value keeps magnitudes strictly positive, so the sign comes only from Stage 1.

Stage 3 — price

$$r_t = +m_t \text{ on } U, \quad r_t = -m_t \text{ on } D, \quad S_{t+1} = S_t e^{r_t}$$

- This is the GBM exponential map with a signed, state-dependent increment instead of $N(\mu\Delta t, \sigma^2\Delta t)$.
- One bar = one trading day in the 1.5-year monthly study ($\Delta t = 1/252$ when a risk-free shift is applied).

Modified GBM • estimation

Closed form on the lookback window. No numerical optimizer.

Laplace-smoothed transitions

$$\hat{P}(U|U) = (n_{UU} + 0.5)/(n_{fromU} + 1)$$

- n_{UU} is the number of up→up pairs among consecutive non-zero returns. n_{fromU} is the number of pairs that start with an up. $P(D|D)$ is the same formula on down→down counts.
- The +1/2 and +1 are Laplace (add-one-half) smoothing so a window with no reversals does not pin a probability at 0 or 1.
- Then $P(D|U)=1-P(U|U)$ and $P(U|D)=1-P(D|D)$. The four probabilities are a two-parameter object.

Split-normal magnitudes

$$\hat{\mu}_U = \text{mean}(|r|_{up}), \quad \hat{\sigma}_U = \text{sd}(|r|_{up})$$

- Down-bar analogues use $r < 0$. If a side has one observation, σ is replaced by the median absolute return in the window. If a side is empty, both μ and σ fall back to that median (or 10^{-6}).
- σ is floored at 10^{-6} so the simulator never draws a degenerate normal.

Rolling window

$$\hat{\theta}_t = \hat{\theta}(\{R_s : t - L < s \leq t\})$$

- L is the lookback (18 months in the 1.5-year 10k study). Update dates are the first trading day of the evaluation window plus each month-end.
- Parameters dated t are used only on steps after t (no look-ahead). If the file is shorter than L , the estimator uses all available history up to t . Equity starts 2003-12-01.

Modified GBM · simulation and pricing

P-measure stock paths · Q-measure LSM calls

P-measure (stock-path report)

- Start every path at S_0 . Initialize the Markov state from last_up in the current parameter row.
- For each step i : read $(P(U|U), P(U|D), \mu_U, \sigma_U, \mu_D, \sigma_D)$ from the rolling schedule at that date. Draw the next sign from the state-dependent Bernoulli. Draw $|N(\mu, \sigma)|$ on that side. Set $r = \pm m$ and $S \leftarrow S \exp(r)$. Do not shift the mean.
- Reported path = 50th percentile across $n_{\text{paths}} = 10,000$. Stock RMSE% is that p50 path versus realized adj-close.

Q-measure (American-call LSM)

- Keep the calibrated transitions and magnitudes. After drawing the P-measure increment r on a step, shift the whole cross-section of paths so that the mean of $\exp(r)$ equals $\exp(r_f \Delta t)$, with $\Delta t = 1/252$ and r_f the 3-month Treasury as of that date.
- The shift is $r \leftarrow r + (r_f \Delta t - \log \text{mean } \exp(r))$. It is a multiplicative drift correction, not a change of the Markov law.
- LSM is Longstaff-Schwartz on the full path cloud ($n_{\text{paths}} = 10,000$, seed 42, $n_{\text{steps}} = \text{DTE}$). Paths are not averaged before the stopping decision. The contract sample is the same Monday ATM listed-call set used by every other model.

Parameter vector written in the estimation PDF

- $P(U|U), P(D|D), P(U|D), P(D|U), \mu_U, \sigma_U, \mu_D, \sigma_D$. last_up is stored for simulation but is not a free parameter; it is the last lookback sign.
- Tables in V3_1p5y_monthly_parameter_estimation.pdf report Mean, SD, Min, Max of the monthly estimates inside each regime and company.

Modified GBM · where the numbers live

Same Monday ATM sample, 18-month lookback, monthly rolling, 10,000 LSM paths.

Empirical ranking (decision)

- V3 1.5-year monthly empirical study 10000 paths / V3_1p5y_monthly_empirical_study.pdf — seven models, four companies.
- Return-based companion V3_1p5y_monthly_empirical_study_return_based.pdf ranks Modified GBM with GBM, GARCH, Merton, and GARCH-Merton.
- Option-implied companion does not include Modified GBM (Heston family only).

Parameters, stock paths, moneyness

- Parameter estimation: V3_1p5y_monthly_parameter_estimation.pdf (Modified GBM table P*).
- Stock paths: V3_1p5y_monthly_stock_price.pdf (P-measure p50 vs S_t).
- Moneyness: V3_1p5y_monthly_moneyness_performance.pdf (same LSM contracts, split by S/K).
- Standalone copies of those four reports also sit in V3 1.5-year monthly Modified GBM/.

Code

- Notebooks: V3-Models_result/modified_gbm notebook/20*_modified_gbm.ipynb. Functions: estimate_modified_gbm, calibrate_ticker, simulate_modified_gbm_rolling.
- Engine: run_v3_1p5y_10k_monthly_modified_gbm.py (standalone cache) and run_v3_1p5y_10k_monthly_empirical_study.py (seven-model 10k cache).